The Free Energy Principle in Our Daily Life With Karl Friston

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hello today i chat with carl fristen who is a computational neuroscientist one of the most cited researchers of all time and we chat about his free energy principle which is has been likened to natural selection as these deeply fundamental principles and ideas to understand how reality works and how it evolves so at carl is this amazing polymath and we go all the way from the big bang through to the future of artificial intelligence and so i hope you listen and check out this episode in order to understand a deeply fundamental principle the free energy principle for how life works and how to kind of see the matrix for what is actually occurring in reality okay thanks and enjoy [Music] hello listeners my name is reese lindmark i'm the founder of root and welcome to the reshow i believe this century is a turning point in human history and i'm here to help you navigate it to do so we'll demystify the hidden but crucial ideas that you can't find anywhere else and by understanding these transformational changes in science technology and society and how to apply those ideas to your daily life i hope to turn you into an active participant in collectively building our solar punk future so with that in mind today i'm excited to chat with carl fristen carl is a theoretical neuroscientist and a professor at university college london he's this amazing polymath who is one of the most sighted researchers of all time carl thanks for being on the show and welcome it's a great pleasure to be here yeah we're excited to dive in and and i've essentially been free energy principal pilled in the last kind of week or so um i've had folks we've had folks on the podcast like andy clark before talking about predictive processing but now i've kind of taken it one step further and fallen further down the rabbit hole and understanding one of carl's crucial ideas this kind of you know free energy principle you've done a lot of work on brains um and now you're kind of you know this this idea the free energy principle some people have described it as the most crucial idea since darwin's natural selection which is a high praise with that as kind of a background and before i kind of give an overview on on some other things we're going to chat about could you just for our listeners carl explain what is the free energy principle what is active inference um and how does it apply to our daily life well if you've had andy on describing predictive processing i think that's a very nice entree into what the free energy principle is about it it's essentially a an attempt to provide a method or a principle of the kind that you would find in physics that would be fit for purpose to understand and explain sentient behavior and of course the kind of scented behavior we're talking about is exactly what andy was talking about in terms of the predictive brain the brain as a a statistical organ that's predicting trying to make sense of its data and then planning and choosing the best way then to go and secure some more data so that's the agenda it's essentially to understand ourselves and how we make sense of our sensations but to do so in a way you can write it down in terms of equations and you can put them to a computer with the agenda then assimilating this kind of sentience and building little artifacts and trying to understand broken brains make more efficient computing devices but of course to do that you really have to become an engineer and a physicist do you know to understand the basic fundamentals so that's the agenda um i could go on and explain the technical foundations of it but perhaps we should save that for your more probing questions later on exactly that's great yeah and i think um as and this is a beautiful part of of carl's work is that and i you know they just released carl along with two other co-authors just released this book on active inference and um a week ago and so i kind of sped read it over the weekend and it is um it's good but it's it's got those formalisms the mathematics in it so it's kind of cool to hear these kind of crucial ideas that do apply for everything but are like connected to formal mathematics that you could actually simulate and code up in a computer with that said let me let me kind of um give you and and our listener some context here for for why i want to touch on the free energy principle and you talked about it in this crucial way is this like understanding of sentient behavior and so you know for me right now i'm writing this book on what information wants which looks at how genes kind of created the tree of life and how memes created this tree of ideas and the free energy principle is you know you know so so my book takes this i kind of big history perspective um on understanding kind of sentient behavior and kind of how these ideas replicate and things like that and so i want to kind of look at you know understanding the free energy principles kind of like touching it from the blind men touching the elephant where you're like what does it look like from this perspective in this perspective and so we're going to take maybe five perspectives today to kind of look at it the first is this kind of information how information flowed in the universe before replicators before genes and memes and we'll look at like whether the free energy principle applies to the sun for example and then we'll kind of go into life and biology and genes and looking at how the free energy principle applies to like the pre-brain organisms like in rna soup well you know what was the free energy principle doing there and then finally you know with memes thinking about both the brain and you know what the free energy principle has to say about the kinds of information that can live in our minds and then also more generally about human society is something like you know these weird like myths that we you know organize around like christianity does that apply or abide by the free energy principle and then finally finally we'll look at machine learning and ai and kind of the combination between free energy principle and um and that so that's kind of what we're gonna do to give you an overview carl and with that let's start with the beginning here with the information pre-replicators so does the free energy principle it's an understanding of sentient behavior does something like the sun comply with it or or how how does it the sun is stable but it doesn't model the environment how do you think about the free energy principle pre-sentient life right so we're going to do everything aren't we so we're starting at the beginning which is excellent yeah um so there is actually a deep connection between um the sun and replication um in the very simple sense that the sun replicates its position um in the sense that it orbits you know around its center whatever that is and of course we orbit around the sun so i use the words orbit or the word orbit deliberately here because that's something that's quite fundamental to the kind of dynamics that we want to understand everything that exists essentially has this aspect that it keeps revisiting states that it has once been in so whether we're talking about the orbits of heavenly bodies uh or whether we're talking about the beating of my heart or whether we're talking about my annual cycle as i look forward to easter my birthday and christmas or whether talking about biorhythms more generally right down to the fast fluctuations and electrochemical potentials in one neuron in one part of my hippocampus at every temporal scale at every level we have this fundamental dynamic where you keep coming back to where you started not unlike a sort of strange loop this is something that is quite crucial and possibly even definitional of things that exist they re-keep their states of being within a limited set so by definition they have to keep on returning to a place in some abstract state space that they once occupied so the free energy principle is then in the game of saying well if things exist in the sense that they keep returning to what's technically called an attracting set of states for the sun it's a very simple well for the earth say it's a very simple set of states it's just that your positions on the orbit around that it pursues the fringy principle then is in the game of trying to understand what dynamical properties must systems possess if they have this natural characteristic but also quite remarkable characteristic that they don't diverge exponentially they don't dissipate they don't decay they don't diffuse into nothing that they keep themselves to within restricted set of states if you like a sort of generalized homeostasis so everything that you want to know about the free energy principle can simply be derived from saying well how must systems flow what kind of dynamics do systems possess if they also possess this this um these attracting um these attracting sets or these orbits if you like um yeah now can i pause this for a second actually sure yeah yeah so that love that and i think it's yeah it's super connected i've heard some people describe the free energy principle as you know if you think about natural selection and survival of the fittest but then you're like oh wait it's actually a survival of the stable you know so you have to have these stable things and in order to be stable then yeah you have um you have to be orbiting around some kind of um attractor set and i think you know for me and trying to understand the free energy principle that visual was so crucial for me it's just thinking about you know like ourselves as beings in non-equilibrium steady states that are kind of you know walking around something you know for for a fish it's good for the fish to be mostly in water most of the time so it can't like go crazy off somewhere else or like for me i like being in cities and you know taking showers and eating food or whatever and so like i'm not just but something like the sun i think is is more difficult because it's it's less of a non-equilibrium steady state and it's also the sun doesn't have a model of its environment and i feel like the model of its environment is a crucial part of three injured principles so tell me how the sun does that you know yep right well in fact the sun doesn't probably um and there's a so that's a really excellent question that you also what differentiates something that could be described i mean clearly the sun's not inert but in the sense of having an active role in the way that it behaves it has a certain inanimacy so i think there's a key distinction between things that exist that keep revisiting exist you know pre-existing states that are animate and inanimate and what you're asking is what differentiates um their inanimate from anime kinds before i sort of drill down on that i just wanted to pick up on the on the you know the lovely connections with evolution and you know just using the word replicator you know uh using the notion of replication immediately talks about this this sort of cyclical this solenoidal just a life cycle is another statement of this fundamental orbital nature so many people would would actually say it is not it is not survival of the fittest that is the best way to understand evolution is just what things persist for long periods of time so it is that persistence that sort of existence in perpetuity for extended periods of time that has this sort of uh stability and interestingly just one final little thing and of course the mathematics of what we are talking about uh the the mathematical definition of a non-equilibrium system or a far from equilibrium system would require it to have a solenoidal flow and what is solenoidal flow and evolution well it's red queen dynamics it's a sort of you know say predator prey light relationship again these continual cycles where you have this your itinerary that has this orbital orbital aspect so your this dynamic is quite fundamental at every level i'm sure it'll it'll it'll appear also in terms of memes that persist at a different level now so let's come back to this um key question um of particular or natural kinds that that do and do not have or look as if they behave as if they have uh some predictive capacity so they're doing some predictive processing or they're engaging in some kind of active inference i think the first thing to um to say as well okay we've defined the kind of system that um that we want to talk about in that it has orbits it has an itinerancy and it occupies a small number of states that it could occupy very much like the officials being in water as opposed to on the streets of new york then um but we've got to do a little bit more heavy lifting here a little bit more work because you know as soon as you talk about something you have to be able to distinguish the states of the thing you're talking about from the states of everything else so you know i have to be able to identify the states of the earth the position of the earth and the position of the states of the sun for example i have to be able to distinguish or disambiguate your states from my states or my states from the rest of the world that introduces this notion of a markov blanket which is central to the free energy principle so on one reading you can look at the free energy principle just as a rehash of quantum mechanics classical mechanics statistical mechanics they're all they all inherit from exactly the same assumptions and with one exception that the free energy principle just puts a markov blanket into the mix and that just enables you to separate the states that are internal to something relative to what's outside and it's that sort of distinction which brings the notion of prediction and information into the game because now you've got a distinction between the inside and the outside where the inside and outside states are separated by blanket states at the surface of the sun on the surface of a cell or your surface or all your sensory and active states and then you can now talk about the inside having beliefs of a mathematical subpersonal sort or statistical um correspondences with the outside and you can now talk in terms of information theory about the inside holding some information or having information about the outside because you've got this sort of exchange across the markov blanket i introduced the notion of a markov planking because i have to do that in order to answer your question about the difference between um the sun and you um the difference um technically it's just the sun doesn't have what's called active states it you know it doesn't have any way of its deep state so let's think about a stone for example um as you know something that certainly exists it's got a uh it's got a mark off blanket or blanket states the surface of the stone you can separate the surface of the stone from everything else and it's got some deep internal states but the key difference between me and the stone is that my internal states express themselves through the blanket they express themselves from the inside out and therefore you can see me behave in a way that looks as if i'm planning i have um you know beliefs about what's going to happen to me that i'm fulfilling um i'm acting in a purposeful way and it may look indeed as if i'm acting in a mindful way simply because uh the my active states are now um in a position to affect the outside again so one simple observation that discriminates a stone from you or a stone from any animate biotic system is that you move so he is as simple as that that you can move in an autonomous way whereas stones and planets they certainly move but in a frame of reference which is you know has no autonomous aspect to it they you know they don't change their shape they're not shape-shifting in the same way that we are and of course if we can move we become animate and we can influence the outside states external states in a way that makes them more predictable basically and you know i'm using that word because you can link that now to andy's formulation you know we are all predictive processing machines and of course if we're if we're good at predictive processing we must be able to engineer the kinds of environment that deliver sensory signals onto our markov blanket that can be predicted and you can get into all sorts of interesting um arguments about niche construction or just being um um just uh the raison d'etre for things like language that we both share that makes you and i mutually predictable and that's how we work but because we couldn't engage in that exchange and leverage that predictability unless we were able to talk and move so you won't see the sun talking to you in in the kind of way that where that we're talking i hope there may be a future in which you know the sun gets it's you know the that's the classic um image of the sun being a like a smiling thing and saying hi good morning you know like we'll have to engineer that into the sun rather than it being natural so i think that your description there makes sense and what i'm hearing there just to reflect it is that yeah that we have these active states that allow us to move and allow us to model and then kind of interact in the environment and one way that i think about it kind of and correct me if i'm wrong here is like there are kind of certain things that can be stable and that can rotate around either non-equilibrium or kind of orbital states in the world and and we label those things objects like sun or a cat or a rock or humans and and there are a lot of those things that are kind of animate and in order to be animate you have to go through this weird like non-equilibrium world and you're kind of you have to have these active states in order to get more energy in order to model the world not just to dissipate but then there's these kind of inanimate ones and those ones you can kind of think of as like the laws of physics have allowed for a certain set of things to exist that are kind of you know the sun just like i'm kind of fit with nice construction you know like a like a fish is fit for its environment so too is the sun kind of fit with the laws of physics in that it can be there's a certain set of stable things that can exist aka a bunch of hydrogen atoms getting together you know exploding making heat you know getting in gravity there rotating around and being a sun and that there's kind of a certain kind of environmental niche that exists that doesn't require active states doesn't require modeling in the environment but is just like an inanimate thing that can exist based off of the uh the laws of the physics is that kind of what do you think about that reflection no i think that's a great picture um and it's particularly um astute because it speaks to this sort of separation of scales both in space and in time and so one way that um some people including myself think of things is that you you as you go from the very very small from the quantum to the um the very very big so we're talking about sort of um you know cosmology and sons and planets and the like then there is a a natural progression from very random behavior to much more precise behavior um and as you get more and more precise and you get bigger and bigger and bigger you actually lose a degree of itinerancy that characterizes you and me i mean we are certainly um i you know characterized by biorhythms and all sorts of fluctuations but the kinds of orbits that we enjoy um are have an itinerancy they have a sort of complexity which is not characteristic of the orbit of the moon for example and that may well simply be um that very very big things don't have the kind of uh itinerancy that would associate with biotic systems so it may well be that um where things like us are are certainly much much bigger than the quantum scale because at the quantum scale then random fluctuations would effectively diffuse or um disperse or mask any of this sort of solenoidal orbital behavior but we're small enough still to have this kind of itinerancy and we only have that eye tendency in virtue of the fact we live in a nice stable world where the moon goes around the sun around the earth and the earth goes around the sun everything uh everything is sufficiently stable over very long periods of time so you know seeing things in that kind of context i think is absolutely essential it is only because we have this slow predictable orbits of the heavenly bodies that are changing the the context on the surface of the earth that provides a simple context in which we can now allow say evolutionary evolution can now emerge and that being a very slow dynamical process that contextualizes the evolution of phenotypes those phenotypes now can learn and what do they learn they learn how to predict and they can predict uh all the way down to the the machinery you need to do that prediction which is basically our brain cells doing that so everything at the finer scale rests upon the right context established by the slower more persistent but still solenoidal still uh orbiting uh scale above but of course it's true the other way around you know you you you you don't get the emergence of uh of species or you don't get evolution unless you have the right kind of phenotype you don't have the right kind of phenotype unless it's got the right kind of brain you don't have the right kind of brain unless you've got the right kind of neuron so at every level this this sort of itinerant orbiting this um persistence in terms of the the dynamics that enable the system to retain or remain in its attracting states depends upon the scale at scale above so that's a real i think a really really important point yeah the free energy principle cannot apply any single level it has to accommodate all of those levels i love that i think i think i love that and i love it as kind of like when people talk about the goldilocks um position for for earth it's like oh we're like not too hot not too cold and that's true but there's a more kind of there's a goldilocks position of scale and there's a goldilocks position of um randomness versus stability you know and so just like there's a lot of special things that need to occur for us to be able to have replicators that can can like learn about the environment and the environment's not being too crazy but it's also not being too stable so yeah that's that's really interesting let's let's kind of take that and move into um moving forward in time from you know the big bang time to now um you know we're starting to get life you know we're starting to get um you know genes and and biology how do you the first question actually is how do you think about the free energy principle there's an abstract question but how do you think about the free energy principle as compared to something like the principles of natural selection and evolution how do they relate to each other which is more fundamental well i think they're both the same so that i that's an easy question to answer and even easier for me to answer in the context of our last exchange about sort of you know how one free energy minimizing process is nested within a higher temporal and spatial scale so technically how i literally think about and how i would sort of articulate this mathematically is i would write down evolutionary dynamics natural selection as a process of what's called bayesian model selection now bayesian model selection is just a a description of what a statistician would do if given a particular statistical model of some empirical experimental data he or she had to optimize the structure of that model say delete a particular parameter from the model or remove a latent state from from the model or if you're in say machine learning it would be uh add an extra layer to your variation auto encoder or remove one or switch out some sort of your rectified linear function for some sigmoid functions there's a structural attributes of anything that can be read as a model then the process of selecting the right model is simply to find the model that has the the greatest or maximizes the probability of the data that it is trying to model that's known as bayesian model evidence aka or also known as marginal likelihood the free energy just is a proxy or an approximation to that marginal likelihood the negative marginal likelihood and indeed in machine learning it's actually known as an evidence lower bound so bound approximation to the model evidence the evidence that this these data or the likelihood that these data could have been generated by this model so if you now think about the likelihood of sensory exchanges in the environment being a measure of adaptive fitness what you're effectively saying is that natural selection is just nature's bayesia model selection of the genetic models namely the phenotypes that have the most likely exchange that i'm going to find when sampling a phenotype at random i'm going to find this phenotype characterized by its sensory exchanges with with this eco niche so from my perspective uh evolution just is an instance of this fundamental imperative to to maximize the the marginal likelihood of the thing as characterized by the states it occupies all the sensory exchanges if i was an evolutionary or theoretical biologist i would call this adaptive fitness um so negative free energy just is adaptive fitness when you take a phenotype and you evaluate the likelihood of its sensory exchanges over its lifetime and then if it's good at what it does and uh it will persist and it will persist in virtue of the fat the natural selection the bayesians the statistician the you know the the great statistician comes along and says yes i like that kind of phenotype that sort of model because there's more of them because they persist therefore they're going to replicate in the next generation so you know natural selection uh is is is for me a beautiful instance of the free energy principle in action but qualified with reference to what we're just talking about this is you know in the context of understanding the coupling between different sort of free energy minimizing processes you know the evolutionary process and the phenotypic or on the ontological process where they both have to talk to each other and you know and that's really the interesting issue is how does the you know the predictive phenotype get selected and how does evolution recognize the the predictive phenotype in a way that is consistent with the free energy principle cool yeah i love that i think that yeah mike it makes me think of this idea of the library of mendel um from like daniel dennett's style book so it's like oh there's all these different um dna strands that could exist and then only a certain subset of those books you know that library do actually exist humans and cats and insects and whatever and so we're kind of choosing a subset of those organisms or those like dna sequences that exist but another way to frame that is it's actually the library of um bayesian models um you know it's like it's a library of like generative models where you're like okay there's a whole lot of generative models that can exist in the world but only some of them are fit with the environment with that kind of multi-scale goldilocks moment and so it's like okay we've chosen these generative models and evolution natural selection is a way of kind of surfacing those generative models within that big old library that actually fit with the environment such that they can do good sensory things such that they can repeat themselves over time um so yeah that makes sense do you think that there's a um you know thinking about other uh biology and free energy principle stuff i think that like i kind of get the brain version of free energy principle it's like okay humans and cats blah blah we got these kind of models of the environment but what about like at the start of life when you have the rna soup and you have these like rna replicators doing their things you know and then oh replication exists and that's a huge moment in universal um time where you have things actually replicating producing themselves in this primordial soup does is the free energy principle at play there or how is it at play when like does like the rna have a model of its environment but it doesn't have a brain so tell me about that right um you've been talking to somebody like mike levin haven't you i i yes yes yes so i'll just reiterate what he would oh no he wouldn't say this because he he's got a beautiful lexicon uh and he's fluent in using it but i'm gonna say what he would say but in maths so yeah a great yeah a great question um so i think you know the principles that we're talking about is just the principles you know the existential imperatives and the principles of things that self-organize to enable them to persist by returning to their attracting set so this has to hold for a thermostat it has to hold for a single cell organism so let's take the single cell i think it's a really nice example it's a particular scale we could talk about molecules we could go up to two brains but let's talk about uh a single cell so i think it's nice that you introduce the notion of sort of rna and dna as basically um nature's specification of the structure majority model because i think that's exactly it in fact you can simulate that quite easily you can just say well this this is the prescription for this phenotypes generative model and this charity model is apt to fit and exchange and explore um this particular eco niche so from a cell i will have a particular organ i live in or a particular medium with the right kinds of nutrients that i will i will seek out if i am sufficiently active so the single cell equipped with if you like at its beginning um scaffolding a specification of the structure in terms of its dna the the most important thing it has to do is to establish its markov blanket it has to be able to maintain um autopoetically its integrity by maintaining its markov blanket it's the the surface of the cell that stops it um dissolving and dissipating into uh into its familiar so even all the electrochemical uh intracellular processes that are in response to cell surface receptors and the expression of active states say the actin filaments actin filaments of a cell which supply with a motility um that could you know be mediated just by movements and morphogenesis or changes in the shape of the cell or it could be have little flagella uh you know little little wings that allow it to uh to move around and then that are that that sort of genetic and epigenetic specification of the morphology of the markov blanket and how that markov blanket the surface of the sensory states are underwritten by active states which cause the right kind of movement that maintains the integrity of the cell these are all just expressions of um the this variational principle least action that just is the the free energy principle what would that look like well if it was quite a sophisticated little cell it would start swimming around where would it swim to well it would swim to those environmental states that were consistent with the integrity of its markov blanket that enabled it to swim so that would look like chemotaxis it would look as if it was searching out or at least it was moving when it was um in you know a concentration um that was either too high or too low in relation to its goldilocks its prior prior preferences encoded by the dna what would happen if a bunch of these cells got together well then we move now to um multicellular behavior and things get a little bit more interesting now uh um you know mike levin tells a lovely story here because you know there's something slightly paradoxical about uh multicellular um collectives in the sense that the the surface cells uh have to sacrifice their ability to replicate so it's you know it's almost sort of anti-evolution but collectively of course they're made just maintaining their mark off blanket then it's persisting in you know in the right kind of way that evolution likes but now the what used to be a markov blanket with its internal external states is now internal to the surface of a multicellular structure so now again we're coming back to this nesting aspect we've got blankets of blankets um so blanket building that your underwrites integrity of a an organelle or a macromolecule structure that then underwrites the integrity of a cellular markov blanket that then underwrites the blanket building capacity of a collection of cells that ultimately grows into a phenotype and a creature and possibly even a society and a species so at every level you are if you like um appealing to the same principles but you can see immediately how something can sort of something's behavior can be evolved or become more and more sophisticated and much more structured in its apparent capacity to plan and to respond adaptively in very particular ways from a little thermostat through to something like you and me or perhaps even beyond through to our favorite political parties or countries or cultures which we'll get to in a second but yeah so that i think that makes sense to me and i think that there's a well hey i love this idea of like you know natural selection kind of choosing it it needs um markov blankets in order to exist and so it's like you know those and those things those blankets whether it's you know lipid layers at you know at very early times or later with like skin cells and stuff it's like you need the the thing itself needs to be able to create that markov blanket in order to have these internal external states and so then and those ones like sacrifice themselves for the good of the whole organism for the good of the whole so that makes some sense to me just think about like the kinds of things that will naturally evolve on any planet it's like you'd want you'd expect markov blanks blankets to evolve wherever um and then the other piece that you said there was just interesting it's like yeah these little kind of you know pre you know like you know you know uh you know survival of the fittest chemical reaction time you know the the very beginning of of of life yeah it's not like they have models but they do have these yeah they have these they have they have generative models that are not like our brains and can't do planning but but i can be in the present and can follow these chemo taxis so i think i think that that makes a lot of sense do you think you know kind of transitioning now into um getting closer to humans and brains we'll kind of you know be in between full-on memes um and and genes with just talking about our brains so one way i think about this is we have this there's this idea of what information wants and the kinds of you know ideas that can replicate from mind to mind and when they replicate from mind to mind they have to have a home in our minds and so something like dun dun dun um can live within our phonological loop you know this little piece of our mind that can kind of like hold audio signals or whatever um and you know so much more about the brain than i do so much more like one could not even say how much more you know but how so how do you think about what kinds of information can replicate to live in our minds like what are the kind of homes that exist in our minds and what information would you expect to replicate and and be stored and be fit for our brains like if our brain was an evolutionary niche what information fits there right so you've introduced two really important themes which i know are close to your heart uh so i'm just gonna unpack them from my perspective the first one um is the difference between um something like a chemotactic cell that looks purposeful it's swimming up the right kinds of concentration gradients and you when you go on google or wikipedia trying to save your knowledge your curiosity and thirst for knowledge and find the right kind of meme uh that's going to populate your your chapter seven of your book and so there's there's a i think there's a fundamental distinction and as usual with um when you think about what distinguishes one kind of thing from another kind of thing or one natural kind from another natural kind it's usually the nature of the implicit generative model and i say implicit because you'll never actually know what goes on the inside because it's behind the markov blanket but it will look as if it's behaving as if it has a generative model and i think the key distinction between the sort of the reflexive in the moment homeostatic behaviors of chemotaptic cells or perhaps even plants um and things like you and me is a temporal depth it's a a depth that allows the genitive model or put more simply the generative model has become sufficiently uh sophisticated that it entertains the consequences of its own action or the system's own action i think it's quite crucial and that brings two key things to the table first of all it now has a generative model of the future so now we're talking about natural kinds that can and cannot plan so you know a thermostat can't plan whereas you and i can plan and that's just because our generative models have acquired a temporal depth that we can actually roll out and predict what would happen if i did that and the second thing of course is the i did that and suddenly now we have a that is all about how this system how i behave the consequences of my actions so there now becomes an implicit autonomy in the sense of agency i'm not saying that all systems autonomous systems know they are agents but it will they will certainly behave as if they have a model of their agency i think that's that's the key thing and then the second big issue you you brought to the table here is you put two of these agents together how are they going to behave so you're talking about sort of language and information well information about what we started off by saying well the free energy formulation puts in a very comfortable position to talk about information in a meaningful way and i mean by a meaningful way i mean information about things not the information of things so we're moving away from shannon-esque information theoretically which is terribly important you know and in a sense the free energy principle is just an information theoretical principle but i i don't think it's kind of information that you want to talk about you're talking about information about things and we're now in a position to say well the internal states of something now can hold information about something else it can be very trivial information could be just some generalized synchrony as you see with your chemotactic cell or your thermostat or it could be information about the future of things out there the consequences of action now come back to the question well what about the universe that has two agents in it well the things out there now are the other agent so now we're in a really interesting situation so both agents are trying to minimize their free energy they're both trying to make their world as predictable as possible and they're going to then find a solution or said evolution will or i should say evo diva or sort of evolutionary psychology will find a solution is a minimum free energy maximum predictability scenario what does that mean it mean well it just means that i am going to engineer my environment to be as predictable as possible or i will evolve and be selected if i can do that what does that mean if my environment is constituted by all sorts of other things like me well it means basically i am going to engineer my environment so that we can all make ourselves as mutually predictable as possible so we're going to have a shared narrative and one could argue um language would be the sort of the most beautiful example of this or certainly a semiotics that is shared between agents so you know it's a a wonderfully challenging and complicated thing to actually think about particularly to model when you have predictive processing systems that have agency so they're inactive they're doing active inference um and they're all trying to predict each other so now the you know i feel like the economy almost disappears and it's just now an ensemble of things that are trying to persist and persevere retain themselves in their attracting states simply by encouraging this shared narrative so i i'm using those words because i think you use the words niche construction and i would i would submit that niche construction just is this slow process of selecting the kinds of phenotypes either by developmental learning or through natural selection at a transgenerational level that underwrite and guarantee a mutual predictability so anything that helps me and you converge on the same generative model of the world and to broadcast that and to exchange that actively through some kind of communication is going to be in the service of making sure that my lived world or my at least my sensed world is as predictable as possible so on that view that shared narrative that generalized synchrony which has now gone beyond just a coupling between the uh the the the total configuration of thermostat and fluctuations the temperature it's now much more complicated all about the future it's all about what are you going to say next what am i going to say whose turn is it to talk now it's your turn exactly beautiful um yeah great ending there i think that there's a i agree with a lot of what you said which is yeah that once we get into these agents that can plan in the future and that those have a sense of themselves as an agent that they know that there are lots of copies of themselves in the world that they need to be aware of those copies and so they know what what their causal loop is connected to like the stuff in the world i was like oh that was done by me i'm the one who's speaking right now once you have all that then yeah the niche becomes instead of just like the environment you know if the sun's niche is the physical laws of the universe and um the inv and then like biological niche for like a cat is like you know the environment a pretty stable thing now we have this weird niche where it's kind of um everything it gets mutual intelligibility and these other agents and so the niche becomes this very complicated thing where you need to have this kind of um yeah this mutuality there and you're trying to kind of in this kind of weird game theoretic evolutionarily stable way you're trying to coalesce around like mutual predictability so that that makes sense to me and and you can see obviously whether it's being mutually predictable with language or mutually predictable with um something like religion or mutually predictable with um you know political parties that all is part of the same kind of idea let me ask let me let me rephrase my earlier question actually though before we get to like religion and things like that which is do you think you know getting down to like the brain level for a second which is because i agree with all your mutual predictability stuff but i want to ask a different kind of question here which is what if we imagine our brain as as this little like home for memes and this home for information you know one way to imagine is from like my simplistic perspective which is these like oh you just like you want sounds like dun dun dun dun to live in the phonological loop and so therefore you know that it'll be inevitable that that thing emerges and that catchy songs emerge and can live in our brains just like it's inevitable that there would be a fish that could live in water but but but that takes a very simplistic view of a brain and you have a much more um uh textured view of what brains actually do and and what generative models they can hold and stuff like that so like what do you think about the kinds of information that can live in brains based on the fact that our brains are these like bayesian brains i think we've already talked about the answer but in a slightly different context so again it's this sort of separation of scales the kinds of things that would um be represented by or constitute the generative model have exactly the separation of a hierarchical sort so that will range from very very low level very fast very small scale representations say of a visual edge in my visual field in a very very small part say just say one or two degrees of visual angle and that will be replicated throughout neuronal representations at the back of my brain and it would just say oh there's an edge orientated in this particular direction at that part of your visual field very elemental um very beautifully engineered but still a very very elemental sort of kind of right up to the very very highest level usually entailing much much greater expanses of time which we've already mentioned in terms of a sense of self selfhood so selfhood is just a hypothesis it's just an explanation that helps me make sense of my sensations my sensorium but it is if you like underwritten by and informed by all this deep processing and that by deep i just mean uh processing undergenerative model that has a hierarchical depth so you've got this notion like of an onion the brain is an onion with all these sensory signals impinging on the outside and all the active states these sort of muscles and autonomic reflexes affecting at a very fast time scale what's going on out there and as you go towards the center of the audience you know to the deeper layers of representation the different kinds information that are represented are meant to be basically those that can predict the unfolding orbits of the layer near the surface lower down or towards the censorium so everything every concept every construct certainly that you could articulate but even those that are so personal you can't articulate like sort of uh having the quality of experience or in fact that's i'm cheating there qualia is another uh another uh representation that you as a hypothesis um but you know from selfhood through to edge detection uh with your eyes or hearing a particular frequency there will be a position in this hierarchical encoding so when you ask me what kinds of uh can a brain represent or encode then they have to be internally consistent in the sense that the the all the different hierarchical levels in this effectively a realization of inference under hierarchical genitive model have to be mutually informing and internally consistent and i think looked out like that then the kind of information that your brain makes space for is just the kind of information that affords a good explanation for your sensed world so everything that makes things simpler and i use the phrase simply deliberately in the sense of occam's principle because um maximizing your marginal likelihood the evidence for your model of the world online in this moment is the same as providing very accurate explanations for your sensations as simply as possible so this is the complexity part of free energy sometimes read as maximum compressibility or minimizing the computational cost or complexity of belief appetiting from our point of view it just means providing a unitary simple parsimonious explanation uh one could contend that the very hypothesis of self is one of these parsimonious hypotheses that provide a good explanation for this remarkable coordination between all my sensory motor proprioceptive and extraceptive sensations hang on the simplest thing is i am a thing and it's me doing all this moving now not everybody has that and certainly newborn babies probably don't have that and certainly lower forms of life don't um but you and i probably do i mean i know i do but i'm assuming that you do i do as well i do excellent so yeah any any idea like that that makes sense of stuff of the kind that when you first realize you're having a har mode oh it's just like that it's it's one of those um anything that has this sort of simplifying uh all all-encompassing explanatory power that allows you to explain away all the free energy the prediction errors lower in your hierarchical journey model is a prime candidate for the kind of information and representation of things out there because these are all you know they're all things in the outside world um i i would submit that's the your that's the that's the way to think about and the answer to the question what kind of information does a brain is a brain designed to um to hold and to commit to beautiful yeah i love that i think my reflection there is yeah that you have these hierarchical structure and at the top level there are these things and i'm reminded of your the book active inference and how there's um continuous signals and continuous generative models and then discrete kind of uh categorical general generative models and what you have to have is for you have to have the things that are fit for our brain are things that that are both fit for the environment that are sensorium that's like oh here's an edge detection and that they're also fit with our like concepts and it has to be mutually fit so it's like yeah you have to have both a thing that's i can't just like make up some random concept that's not actually fit with the environment and i also can't make up a um random like a continuous like stream of sensorial information that doesn't have any concept attached to it it needs both in order to be fit for to our mind and something like the self or a rock or whatever is a thing that is actually what's out there in the world and we kind of label it as this kind of categorical thing to it um so that all makes a lot of sense i have a a question for you carl which is um is it okay we're going can we go like 10 like if 10 or 15 minutes over time right now or do you have uh something else to do please carry on okay great that'd be great yeah because i think there's a couple crucial things that we'd love to hit um right now that i'd love your thoughts on so first is let's go to the um so now we understand the brain um and we understand genes and we understand uh the big bang and now we're going to um this higher level these kind of nested markov blankets that you were chatting about before which is you know this these larger social structures and kind of society um and so there's kind of maybe a two-part question here which is like how does you know the free energy principle apply to these social systems and i know you're talking about it a bit before something like mutual um intelligibility and that kind of like thing but also maybe like a specific example just like we use the sun as a specific example before does something like christianity um you know christianity is both the symbol it's the cross it's this kind of group of people it's this set of ideas and concepts does something like christianity is that like abiding by the free energy principle does it have a model of its environment and it's a model of its environments just like these humans and the brains and stuff like that so so how do you think about how the free energy principle applies to like social systems very challenging an excellent question i wouldn't say that christianity as a construct has a generative model but certainly um ensemble an ensemble of agents who are engaging in active inference should come to behave if they're sufficiently curious and sophisticated and um mathematically empathetic to the the other members of the ensemble they should show the sign the kind of community uh communicative exchange and behavior which would have all the hallmarks of something like christianity you might also expect that given a sufficiently large sort of cultural landscape that you get sort of 50 50 splits so you'd expect sort of to have another kind of religion which was not christianity and you'd normally expect there to be about 50 50 splits in a in an evolutionary stable um steady state sense um and there are good and principled reasons for this and even if they're not easy to intuit mathematically you can easily demonstrate this using numerical analysis or multi-agent games the the the the emergence of this is is very interesting um and it comes back to something we mentioned before um about the difference between um animate objects that have this sort of reflexive aspect uh and the very very simple versus sentient creatures like you and me with a sense of agency and these deep generative models that free us from the moment because we we can imagine the consequences of our action in imagining the consequences of our own action we have to now plan what we're going to do we have to have um a notion and narrative so it's no longer just oh i'm going to swim in that direction or this direction or i'm going to switch the radiator or off because the temperature has dropped it's much much more a much more deeply temporal structured navigation of the world with narratives but narratives have to be chosen so we're talking about planning as inferencing we're talking about creatures that can plan and ensembles creatures that can plan if they plan they have to have the ability to score the goodness of one plan or the likeness of one plan relative to the other and it turns out that mathematically those plans whose outcomes or consequences have the smallest expected free energy are those that have the greatest information gain what does that mean well it means that anything that has a deep generative model that exists must therefore behave as a curious creature now if you're a curious creature you've got a problem because you now have to select the kind of data that's going to resolve your uncertainty so you have to be able to estimate what kind of data what kind of exchange for example is going to have the greatest expected information game which is an important part of the expected free energy so you now get into this very difficult situation which we all contend with which is who do we listen to what news channels do we listen to what papers do we read what social media do we commit to which church do you attend on sundays these are all choices we make that contextualize and have a profound effect on the sensory evidence for our genitive models and so when you put this into the mix then the obvious thing of course is that for those things that are sufficient like me um and i have direct exposure to then we're all going to converge on the same constructs the same narrative the same uh cultural commitments in terms of the language we use the social norms the way that we uh make things easier for ourselves in terms of making things predictable with deontic cues and signs gestures um you know in language um you know you can see evidence that much of our communication is just basically pointers you're like me i'm like you so uh you know let's start our engagement uh however in so doing then if there's somebody in another group who is now infers ah well that's not my group but i'm i'm in this group they are going select those policies that they think are going to have the greatest information again resolve their curiosity and make everything um internally predictable so they can learn their environment but it's not going to be the same as yours and usually the only stable strategy is you've got a 50 50 split because if one's too small it's not getting it's absorbed into the big one so the only stable one is is christianity versus not christianity or brexit versus not brexit or trump versus biden wherever you look you know these sort of cultural commitments that enable you to seek out the kind of information that reassures you yes you know this is the kind of thing that you are um will uh will inevitably involve some segregation so now you you actually got another instance of blanket building now you've got a segregation which is a completely functional and necessary part of self-organization you know it's autophagys self-assembly defined operationally by constructing building blankets and preserving them to isolate to segregate um in a in a way that defines not only you as an individual but your uh kin your culture your species uh your place in in in this world that's going to um you know be a vince in this instance at the level of cultural or social norms so another beautiful example of the on the same kind of blanket building principles this auto police's self-assembly in chemistry this would be self-assembly in cellular biology it would be waterprooesis um i'm sure there's a good word for it in in cultural psych evolutionary psychology you you do you know what the word is what is the tendency to segregate yeah i mean it's a good question i i'm thinking about there's like yeah group that's a good question i know of um i don't know actually what that would be called um you know group identities um identity construction something like that is is are the things yeah i'm not sure yeah yeah but we'll i'll google it and get back to you but yeah it's that it's that same like you said it's just like you know no matter what um the universe uh needs it it evolves towards blanket building and this is just another form of blanket building is the tendency to segregate and and as you say it's like yeah and the crucial part of it is that you want as we move through our lives to um yeah to have things that are are are stable and are clear and that the information that we're getting is is not all crazy all the time but it's like stuff that we know and then we feel safe with so that that makes a lot of sense to me and i think i mostly agree with you on christianity not having a um a model of and it obviously doesn't have a generative model in the same way that we do but there's something about christianity in that it knows how brains work it it has evolved to be this thing that does know because if it didn't then it would it would die out and so there is some sense that has a generative model do you kind of see what i'm saying no sorry yes you're absolutely right in in that sense in the same sense that we that me as a phenotype is a you know i am an embodiment of the generative models of my eco niche you're absolutely right and i think christianity as a construct is exactly that kind of thing what i meant was it may not have a deep judging model it doesn't plan um so as with many things that get big they that they don't plan quite so much anymore uh like the sun doesn't which is you know or the um the you know a cosmological scale you lose the ability to plan so it's a it's a similar issue a question you know does does natural selection plan uh and i think it would be difficult to argue in the sense that you and i plan that natural selection plans but certainly natural selection i'm sure i'm absolutely sure that you're right in relation to sort of um things like christianity um that you'll have a well-defined markov blanket then they certainly have a generative model i'm just saying it may not be a temporary deep generative model no i agree with that i think i think there's an interesting it makes me think of um and this is this world of memes and thinking about how there's some memes that have a super that um like virals that just spread very quickly and so this is like some kind of hip new thing but then i will say that evolutionarily there is something about something like christianity it doesn't have it it can't plan but it does a workaround for planning which is that it it gets access to long-term energy so it gets um lots of bequests to the church so that it can get more money or something like a corporation that has like you know apple has a really big balance sheet and has a lot of money um kind of like how the sun has lots of access to energy so it can perpetuate for billions of years or whatever so too do these like longer term i agree that doesn't have the ability to plan but it has something within it that allows it to retain for longer such that when new information comes in it like doesn't just it doesn't just go away like a viral does yes so um let's go let's move to our fight ago well no i'll just say i mean but you've just you you i think you've beautifully articulated that you know the existential imperative just to persist you know it's last man standing and of course you also say it has to have the right context in which the last man can stand uh you know so it has to have that as how that resource it has has the sun without the sun there might not be any christianity because there might not be any others exactly exactly um and that can take the form of uh you could be last man standing by knowing what's going to happen in the future or you can be last man standing by just having a reserve so that weird things happen in the future you're like ready for it and you're kind of your meme or your replicator can kind of work through the new environmental niche that exists um so moving to our final question here um and and thank you again carl for the bonus time so talking about so we've we've gone from genes to memes and now we're in this weird new reality where we got computers and um you could call them you know susan blackmore calls it like the new third replicator and you know she calls them dreams i like to call them keems like these computer memes um and so there's there's something weird here about this new kind of thing that's emerging and a lot of folks are worried about ai safety and how this thing is kind of maybe optimized for some things for different values than what we're optimized for and and that we might just kind of you know get turned into paper clips or whatever so how and i know that the free energy principle has a lot of overlap and you've kind of discussed some of it with machine learning and things like minimizing the elbow and minimizing kind of loss functions and stuff like that how so so i guess there's a two-part question here one of which is like how does the free energy principle apply with machines and ai having kind of generative models of the world and then maybe the second question connected to that is like how then does um what does that imply for ai safety and like how we should be making sure that its generative models are ones that include us and that like don't want us to die or whatever i i think that the you know sort of certainly the active imprints instantiation the free energy principle has a lot to say about artificial intelligence and generalize artificial intelligence um in a way that moves us beyond reinforcement learning applications or deep rl applications and i think what it has to offer is exactly uh what we've been talking about in terms of this artificial curiosity so if you haven't had him on yet then yoga schmidt hoover would be the man to wax lyrical about about the importance of sort of creativity fun and curiosity which is a you know this sort of information gain uh novelty seeking uh aspect of um expected free energy minimization what we're looking at in terms of the next generation of computer artifacts or edge computing or robots are basically curious robots you'll now have intelligent data mining because part of curiosity is knowing where to look so you know i suspect there's going to be a this is probably rubbish i'm talking rubbish until about five or ten years have passed but i'll say it anyway my guess is that in the next five to ten years big data is going to die and what will be replaced with it will be replaced with tiny sparse salium data so it'll be all about the quality of the data mining searching out the right kind of data that has the greatest information gain that's going to resolve the uncertainty that this data miner this robot or this or this user so that's going to be i think a change in the future which of course if you want to bring in sort of climate change issues and the amount of energy we currently spend on on computing is quite uh is quite an important argument from the point of view of sustainability but i think mathematically that's what you're active in for the free active inference and the free energy principle uh brings to the table and it also relates this minimum complexity maximum compression argument that the the most likely artifacts including computers that will exist in the next decade or so are those that are most efficient and have the most simple way of providing accurate predictions in the simplest way possible so they will be cheap they will run on batteries you won't need generators to do your uh to do your uh to do your date of mind because you'll you'll know where to look what the in terms of sort of you know the the paperclip um issue i think that that inherits from um a behaviorism um and a commitment to reinforcement learning and value function um formulations of optimal behavior that kind of formulation doesn't exist in the free energy principle you know there is no value function or reward function all there are are preferences about the outcomes of behavior and those preferences are just the kinds of things that are you are most likely encounter or produce as a consequence of your behavior and as such um if you produce an outcome that makes somebody cross like making too many paper clips or accidentally starting world war three then you're not gonna see that you're only gonna do that once and it's not gonna be the kind of thing you prefer producing what i'm trying to say here is built into the normative account ascension behavior is a safeguard against the the paper clip manufacturing automaton because it's a whole point is to make its world as predictable as possible so it knows what the likely outcomes are because it can predict what those outcomes are and those outcomes as we've said about four times now are deeply contextualized by the environment and the eco niche in which this agent is operating so what that means from my point of view is that it's more likely that the next generation of artificial intelligence will take its lead from um automated pets and carers so you know the things that that are going to sell that are going to be bought off the shelf and therefore fund research and incentivize people applying the next level of machine learning which one might submit would be active are going to be um sort of curious artificial pets sort of series that are really interested in how you're feeling they're curious about you and they're gonna ask you because they want to know about you because it's your that you're their so this is a much more benign picture i think where there's where you actually a co-evolution of artificial and natural uh intelligence um in the same way that we have a co-evolution with our with our pets then we have a coevolution with our agriculture and with with our technology um so i i don't think that i can't imagine provided that the free energy principle is an apt description of our planetary evolution i can't imagine um you know things the singularity that you you see popularized in science fiction and some of the popular science literature yeah beautiful yeah i think that there's um so two pieces there one is yeah i love the idea and i agree with it to some extent which is like right now we're just doing brute force just like you know give the gpt3 just like 70 billion parameters and 96 layers of the transformer just let it roll you know and actually and even even transformers this new kind of artificial intelligence architecture it does this kind of what to look at thing it has this like list of like what you know when you're looking at text it has a um an indicator for where in the past it's looking and the layers kind of tell it where in the past to look to understand to predict the next um kind of uh word and the sentence or whatever and so it's it's our we're already starting to do this kind of look only at specific things in the past and then like you say i also agree with like and we're already exploring this it's like we've done the big data thing but of like oh just more data equals better models and actually what they're finding now is there's this new one where they all they asked it was to they um instead of like a 70 billion parameter model like at gbt3 it was like i think a seven billion parameter model that outperformed it but what it did is it they asked it hey ex like explain your reasoning as you're doing it um and it got it for some reason that just made it so much better and so i just i just think that you are probably directionally correct and from a free energy principle perspective it's like finding out the kinds of information game the kinds of feeds just like we have our own kind of feeds to that we consume from what feeds will it be doing and not just all of them but specific ones um to to go for so i agree with that and then on the side of the ai alignment yeah i think i mostly agree with that i think that there's a and there's this kind of like co-evolutionary perspective that i think a lot of the ai safety people don't take into account they're just like oh no value function and the value function gets optimized for and we're all going to die it's like okay wait there is this co-evolutionary domesticated relationship where we domesticate each other we're both trying to mutually co-understand each other and mutually come to this like shared world where we both exist as like agents in this world and i think the hope would be you know when they do bad things um that we are that's not a fast enough takeoff scenario such that when they do the bad thing we say bad and then their gender model changes that they're like okay i won't do that again and we hope that when they do it bad the first time it's not it doesn't like ruin everything but just like only has a small impact so with that said thank you so much carl for the the convo today i think i learned a lot and hopefully the listeners learned a lot um as a note where should people i mean hey check out i just read carl's carl is a prolific extremely prolific which is really cool um check out my recommendation i'll add these in the show notes i'll add some kind of like um things that will help you you know so like this amazing 15-minute video that carl did about getting you into free energy principle i'll also link to the new book about active inference if you want like a deeper dive especially around how brains do it in the kind of a more formal kind of mathematical sense and bayesian sense um carl do you have anything else to that where our listeners should go or any other things you'd like to tell them thank you for the opportunity but no i think i think you did a grand job of everything that will be useful thank you very much beautiful um well with that listeners thank you so much for coming um and goodbye everybody bye carl thanks so much for listening today if you like the show please give us a 5 star podcast review or subscribe on youtube and if you'd like to chat about this episode with a community of amazing smart ambitious divergent people come on by and join our discord you can find it at root.co that's r-o-o-t-e dot co and then finally if you'd like to contribute to these ideas being shared more widely in society you can support the podcast production team at patreon.comreeselandmark that's patreon.com r-h-y-s-l-i-n-d-m-a-r-k thanks and see you here for the next episode bye